PI-THERMAL: the label printer

Supply

J1
USB-C trigger 12 V
+12V
USB-C
OUT+
OUT-

U1
M5 printer DC-DC
+12V
+5V
IN+
IN-
OUT+
OUT-

J2
Pogo (3D print)
+5V
5V
D+
D-
GND
USB_DP
USB_DM
GND

U2
Raspberry Pi Zero 2 W
+5V
TP_5V
TP_USB_D+
TP_USB_D-
TP_GND
2_5V
1_3V3
15_GPIO22
16_GPIO23
18_GPIO24
19_GPIO10_MOSI
22_GPIO25
23_GPIO11_SCLK
24_GPIO8_CE0
6_GND
37_GPIO26
REV
GND

U3
Adafruit Mini PiTFT 1.14"
5V
3V3
BL
BTN_2_Up
BTN_1_Enter
MOSI
DC
SCLK
CS
GND

Header 2x12 (pins 1–24), SPI0.
Buttons: gpio-key overlays (tui.py).

The printer (M5)

U4
M5 printer board
+12V
VIN_12V
GND
USB_D+
USB_D-
MOT_A1
MOT_A2
MOT_B1
MOT_B2
OUT_A1
OUT_A2
COIL_A1
COIL_A2
USB_DP
USB_DM
M1
Paper feed

USB: the custom Mini-B plug, D+/D- straight to the pogo pins.

Planned: reverse feed (DNP)

U5
MAX4624
+5V
C1
100n
V+
COM
IN
NC
NO
GND
COIL_A1
REV
OUT_A1
OUT_A2
R1
10k
GND

U6
MAX4624
+5V
C2
100n
COIL_A2
REV
OUT_A2
OUT_A1
GND

Planned (dashed): REV high = COM to NO on both switches: coil A reversed, so the same blank raster rows feed backwards. R1 keeps it forward while the Pi boots. Switch only with the motor still.
MAX4624: 1 ohm, 200 mA cont. / 400 mA peak, COM/NC/NO within 0..V+ (5 V). To check first: the driver's motor supply <= 5 V and the coil current; if not, the switches go on the driver's inputs.
Today coil A is wired straight: OUT_A1 = COIL_A1, OUT_A2 = COIL_A2.

pi-thermal

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Size: A5	Date: 2026-10-18	**Rev: 1**
KiCad E.D.A. 9.0.8		Id: 1/1